Bone broth: HMTc Standard

Karen Pendergrass · 0000-0002-2348-7259 · Heavy Metal Tested & Certified, Institute of Contaminant Standards

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CORRESPONDENCE

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KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

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Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for bone broth — long-simmered animal-bone and connective-tissue stock (Prepared and Packaged Meals), evaluated as consumed. Each limit is a disclosed read-across from the strictest credible-government maximum level for the product's principal animal components: lead 100 (FSANZ meat), cadmium 50 (EU meat), total mercury 50 (Hong Kong Cap 132V meat of animal). Where no government regulates a component the analyte is a mandatory control, and the species cells (inorganic arsenic, methylmercury, hexavalent chromium) are substantiated not-material determinations bounded by their measured total.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default; COMPOSITE — disclosed read-across from strictest applicable principal-component sovereign ML; never occurrence-derived)

2 | Master limit table

All values in ppb (µg/kg), as consumed basis. A brand meeting every value qualifies for the mark.

ANALYTE	HMTC LIMIT	BASIS
Lead (Pb)	100	human-grade read-across ()
Cadmium (Cd)	50	human-grade read-across ()
Total arsenic (tAs)	500	human-grade read-across ()

Inorganic arsenic (iAs)	n/m	not material in this matrix
Mercury, total (tHg)	50	human-grade read-across ()
Methylmercury (MeHg)	n/m	not material in this matrix
Nickel (Ni)	control	measured every lot; no human-food maximum
Tin (Sn)	control	measured every lot; no human-food maximum
Aluminium (Al)	control	measured every lot; no human-food maximum
Total chromium (Cr)	1,000	screening ceiling (Hong Kong)
Hexavalent chromium (Cr-VI)	n/m	not material in this matrix

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	100	— human-food analog
Cadmium (Cd)	50	— human-food analog
Total arsenic (tAs)	500	— human-food analog
Inorganic arsenic (iAs)	n/m	There is no inorganic-arsenic source in a terrestrial-animal bone broth: no rice, no marine bioaccumulation, and no arsenical-treated plant component. Inorganic arsenic is bounded by the PUBLISHED total-arsenic ceiling of 500 ug/kg on the tAs cell (iAs <= tAs), and reflex speciation is triggered rather than waived. Fish-bone broth (rare) inherits the Category 6 seafood inorganic-arsenic ceiling.
Mercury, total (tHg)	50	— human-food analog

Methylmercury (MeHg)	n/m	Methylmercury has no formation pathway in a terrestrial-animal bone broth and is a component of total mercury (MeHg <= tHg). It is bounded by the total-mercury cell. Only a seafood-derived bone broth carries a methylmercury pathway, and that inherits the Category 6 seafood ceiling.
Nickel (Ni)	control	No credible regulator sets a nickel maximum level for meat, bone, offal or animal-derived stock. Instruments checked: Commission Regulation (EU) 2024/1987 amending 2023/915, whose nickel maximum levels are plant-matrix only (fruiting vegetables 400, brassica 500, leafy vegetables 500, fruit/vegetable juices and nectars 250, cereals 800, root/tuber/bulb vegetables 900, legume vegetables 1000, pulses 4000, chocolate and cocoa products 2500 to 7000 ug/kg) plus infant-scoped rows that are Category 1 scope; EU 2023/915 itself (no nickel entry before the 2024/1987 amendment); Codex CXS 193 (no nickel entry); FSANZ Schedule 19 (no nickel entry); Hong Kong Cap. 132V Part 2 Schedule, whose only nickel rows are packaged-drinking-water and natural-mineral-water parametric values (70 and 20 ug/kg) in a different basis family that this row's basis_guard refuses; US FDA (bottled-water parametric value 100 ug/kg only); Health Canada and UK dietary instruments (no nickel ML). v1.2 CORRECTS A CITATION ERROR: v1.1 attributed the nickel maximum levels to 'EU 2024/1252', an unrelated instrument. Every nickel row in the register is Commission Regulation (EU) 2024/1987 (regulation_page regulations/eu-2024-1987--nickel-maximum-levels). The substantive conclusion is unchanged and was independently re-verified: every credible nickel food row is plant-matrix, infant formula, baby food or chocolate, so none reaches an animal-derived stock and no near-neighbour read-across is supportable. Nickel is therefore a mandatory reported cell with no invented ceiling; claim analyte. INGREDIENT-WATER RULING, and a correction to what the basis_guard can actually do. The Cap. 132V packaged-drinking-water and natural-mineral-water rows sit in the SAME basis string as Cap.

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Tin (Sn)	control	Inorganic tin is a tinplate migration pathway confined to the canned form; carton (RTD), frozen and refrigerated bone broth have no tin source, so no ceiling is asserted for them. A CANNED bone broth is verified against the canned-foods tin ML, on which three credible regulators converge at 250 mg/kg (250000 ug/kg): Codex CXS 193 (canned foods), FSANZ Schedule 19 (all canned foods) and HK Cap. 132V (canned food other than canned beverages). Instruments checked for a non-canned form: EU 2023/915 (tin appears only as canned-food and canned-beverage MLs), Codex CXS 193, FSANZ Schedule 19, Cap. 132V, US FDA, Health Canada and UK dietary instruments; none sets a tin maximum level for a broth outside a can. The Cap. 132V canned-beverage row (150 mg/kg) is stricter but is not adopted: a ready-to-drink carton broth is not a canned beverage and has no tinplate contact. THE STRICTEST APPLICABLE CANNED-FOOD TIN ML IS THE EU ROW AT 200 mg/kg, NOT THE 250 mg/kg FIGURE v1.1 NAMED. Commission Regulation (EU) 2023/915 sets tin at 200000 ug/kg (200 mg/kg) for canned foods other than canned beverages, except canned dried and canned powdered products (eu2023_sn_canned_foods_-200000, basis product as placed on the market). It is stricter than the three rows v1.1 named as converging at 250 mg/kg (codex_cxs193_sn_canned_-250000, aunz_sch19_sn_canned_250000, hk_cap132v_sn_canned-food--other-than-canned-beverages_-250000), it is a credible-regulator maximum level in a basis this row accepts, and it was neither named nor dispositioned. This defect was found by hand during the v1.2 remediation and NOT by the compiler safety-net, because the soups row's audit_exclude_scope contained the term beverage, which matched the phrase 'canned beverages' inside the EU row's own scope string and suppressed it from every scan; the same class of regex self-defeat the v1.2 audit identified elsewhere. The term is removed. The canned form is verified against the EU

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Aluminium (Al)	control	No credible regulator sets an aluminium maximum level for bone broth, for any animal-derived stock, or for any animal-tissue component of one. Instruments checked: EU 2023/915 and EU 2024/1987 (no aluminium entry), Codex CXS 193 (no aluminium entry), FSANZ Schedule 19 (aluminium appears only as infant-formula MLs of 500 and 1000 ug/kg, which are Category 1 scope), Hong Kong Cap. 132V Part 2 Schedule (no aluminium entry), and US FDA, Health Canada and UK dietary instruments (no aluminium ML). Aluminium reaches the credible register only in basis families that cannot anchor a food-content cell: drinking-water parametric values (EU 2020/2184 and US 21 CFR 165.110 at 200 ug/L, Health Canada at 2900 ug/L), food-contact-material specific migration limits (EU 10/2011, a migration figure into a simulant), and toy migration limits. Permitted-additive maxima for aluminium-containing additives exist only in non-credible-tier instruments and are additive-use levels rather than contaminant ceilings. Aluminium is therefore a mandatory reported cell with no invented ceiling; claim analyte.
Total chromium (Cr)	1,000	government maximum level
Hexavalent chromium (Cr-VI)	n/m	Hexavalent chromium is reduced to Cr-III in a cooked, wet, protein-rich broth; a Cr-VI result is an artifact of the total-chromium screen. No credible regulator sets a Cr-VI food ML.

4 | Cite this standard

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STATUS AND LICENCE

COMPETING INTERESTS

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

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DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/bone-broth. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

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